

Stabilizers for $\mathrm{GL}(V)$ on 2-dimensional subspaces of $\wedge^2 V$ when $\dim V \leq 7$

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Abstract

We study the action of $\mathrm{GL}(V)$ on $\mathrm{Gr}(2, \wedge^2 V)$ for $\dim V \leq 7$. Using weak-congruence labels only as names for the chosen representatives, we record the geometric orbit representatives together with the exact stabilizer sequence

$$1 \rightarrow K_L \rightarrow \mathrm{Stab}_{\mathrm{GL}(V)}(L) \xrightarrow{\rho_L} Q_L \rightarrow 1$$

for every row over an algebraically closed field. The row-by-row geometric kernel and quotient calculations are verified in a companion Lean 4 formalization, while the finite-field descent and orbit-count identities are checked with companion Magma scripts. The finite-field tables contain exactly 4, 6, 14, 20 orbits for $n = 4, 5, 6, 7$. Summing the corresponding orbit sizes gives the whole Grassmannian $\mathrm{Gr}(2, \wedge^2 \mathbf{F}_q^n)$, so the finite-field tables are complete. Consequently the numbers of geometric orbits are 3, 5, 11, 16.

1 Introduction

Let V be an n -dimensional vector space over a field k . In this paper we study the natural action of $G = \mathrm{GL}(V)$ on

$$\mathrm{Gr}(2, \bigwedge^2 V).$$

Equivalently, we study 2-dimensional subspaces of $\wedge^2 V$ up to the action of $\mathrm{GL}(V)$.

This problem lies inside the general finite-orbit theory for algebraic groups on subspace varieties. Applied to the $\mathrm{GL}(V)$ -module $\wedge^2 V$, the results of Guralnick, Liebeck, Macpherson, and Seitz [GLMS97] imply that there are only finitely many $\mathrm{GL}(V)$ -orbits on $\mathrm{Gr}(2, \wedge^2 V)$ for $\dim V \leq 7$.

There are two nearby pieces of literature. First, the weak-congruence classification of alternating pencils due to Sergeichuk and to Drozd–Plakosh provides the orbit types in principle [Ser10, DP14, DP16]. Second, the papers of Brooksbank–O’Brien, Brooksbank–Wilson, Brooksbank–Maglione–Wilson, and Brooksbank–Li–Qiao–Wilson develop algorithms for preservers, isometry groups, and pseudo-isometries of alternating matrix spaces [BO08, BW12, BMW17, BLQW20]. What is still missing from this picture is a uniform exterior-square orbit table through the finite range $\dim V \leq 7$: explicit wedge representatives, their geometric stabilizers, and the resulting finite-field orbit tables. This is the gap filled here.

For a representative $L \leq \wedge^2 V$, let

$$\rho_L : \text{Stab}_{\text{GL}(V)}(L) \rightarrow \text{GL}(L) \cong \text{GL}_2$$

be the induced action on the 2-space L . The paper records

$$K_L = \ker(\rho_L), \quad Q_L = \text{im}(\rho_L),$$

so that the stabilizer is described by the exact sequence

$$1 \rightarrow K_L \rightarrow \text{Stab}_{\text{GL}(V)}(L) \rightarrow Q_L \rightarrow 1.$$

The row-by-row geometric calculations establishing these kernel and quotient descriptions are verified in a companion Lean 4 formalization [Riz26]. The finite-field descent and orbit-count identities are then checked in the paper, with companion Magma scripts used for the symbolic orbit-count simplifications.

Weak-congruence labels are used only as names for the chosen representatives. They are not used to determine the quotient Q_L . Instead, Q_L is computed directly as the image of ρ_L . We keep row numbers for easy navigation in the tables and in the Lean crosswalk.

In the stabilizer formulas, U_n denotes a connected n -dimensional unipotent group, and $U_n(q)$ denotes its group of $\mathbf{F}_q$ -points. When the unipotent radical is explicitly a vector group, we write $\mathbf{G}_a^m$.

The main result is the following.

Theorem 1.1. *Let $k = \overline{\mathbf{F}}_q$, let $V = k^n$, and assume $n = 4, 5, 6, 7$.*

1. *The geometric $\text{GL}(V)$ -orbits on $\text{Gr}(2, \wedge^2 V)$ are represented by the weak-congruence rows listed in Section 3.*

2. *For each row L , the algebraic stabilizer is determined by the exact sequence*

$$1 \rightarrow K_L \rightarrow \text{Stab}_{\text{GL}(V)}(L) \rightarrow Q_L \rightarrow 1,$$

with K_L and Q_L listed explicitly in Section 3. These geometric stabilizer calculations are verified row by row in the companion Lean 4 formalization.

3. *Over $\mathbf{F}_q$, the geometric families refine to exactly 4, 6, 14, 20 finite-field orbits. Their stabilizers are listed in Section 4.*

4. *For each $n = 4, 5, 6, 7$, the sum of the orbit sizes determined by those finite-field stabilizers equals the number of 2-spaces in $\wedge^2 \mathbf{F}_q^n$. Therefore the finite-field orbit tables are complete. Consequently the numbers of geometric orbits are 3, 5, 11, 16.*

The companion repository contains the Lean 4 formalization of the geometric stabilizer calculations and the Magma scripts used for the finite-field tables and orbit-count identities. It is available at <https://github.com/alunik/gl2spaces-wedge2>.

Section 2 fixes notation and the weak-congruence naming convention. Section 3 gives the geometric orbit and stabilizer tables together with the Lean crosswalk protocol. Section 4 carries out the finite-field descent, records the finite-field orbit/stabilizer tables, proves the orbit-count identity, and deduces the completeness of the geometric list.

2 Preliminaries

Let k be a field and let V be an n -dimensional k -vector space. A point of

$$\mathrm{Gr}(2, \bigwedge^2 V)$$

is a 2-dimensional subspace $L = \langle \omega_1, \omega_2 \rangle$ of $\bigwedge^2 V$, equivalently a pencil of alternating forms $\lambda\omega_1 + \mu\omega_2$ on V .

The action of $G = \mathrm{GL}(V)$ on $\bigwedge^2 V$ induces an action on $\mathrm{Gr}(2, \bigwedge^2 V)$. For a representative $L \leq \bigwedge^2 V$, define

$$\rho_L : \mathrm{Stab}_{\mathrm{GL}(V)}(L) \rightarrow \mathrm{GL}(L) \cong \mathrm{GL}_2$$

to be the induced action on the coefficient space L . We write

$$K_L = \ker(\rho_L), \quad Q_L = \mathrm{im}(\rho_L).$$

Thus K_L is the pointwise stabilizer of L , Q_L is the image of the setwise stabilizer on L , and the stabilizer is organized by the exact sequence

$$1 \rightarrow K_L \rightarrow \mathrm{Stab}_{\mathrm{GL}(V)}(L) \xrightarrow{\rho_L} Q_L \rightarrow 1.$$

The geometric tables record the chosen representative together with the corresponding pair (K_L, Q_L) .

When k is algebraically closed, we index the rows by their weak-congruence type. These labels are used only as names for the representatives. The quotient Q_L is computed separately as the image of ρ_L ; it is not determined from the weak-congruence label alone.

For $0 \leq r \leq \lfloor n/2 \rfloor$, set

$$\omega_r = e_1 \wedge e_2 + \cdots + e_{2r-1} \wedge e_{2r}.$$

We use without further comment the standard classification of lines in $\mathbf{P}(\bigwedge^2 V)$ by even rank and the corresponding stabilizer descriptions for the model tensors ω_r . These background facts serve only to fix notation and to connect the chosen row labels with the existing weak-congruence literature.

The finiteness theorem of Guralnick–Liebeck–Macpherson–Seitz [GLMS97] shows that $\mathrm{GL}(V)$ has only finitely many orbits on $\mathrm{Gr}(2, \bigwedge^2 V)$ for $n \leq 7$. Those are precisely the dimensions tabulated in Sections 3 and 4.

3 Geometric Orbit and Stabilizer Tables

For this section let $k = \overline{\mathbf{F}}_q$ and $V = k^n$ with $n = 4, 5, 6, 7$. The following theorem records the geometric orbit representatives together with the exact kernel and quotient data attached to their stabilizers.

Theorem 3.1. *For $n = 4, 5, 6, 7$, the $\mathrm{GL}(V)$ -orbits on $\mathrm{Gr}(2, \bigwedge^2 V)$ are exactly the rows listed in the geometric orbit tables below. For each representative L , the stabilizer is determined by the exact sequence*

$$1 \rightarrow K_L \rightarrow \mathrm{Stab}_{\mathrm{GL}(V)}(L) \xrightarrow{\rho_L} Q_L \rightarrow 1,$$

with K_L and Q_L given by the corresponding row of the geometric stabilizer tables.

Proof. The kernel and quotient statements are verified row by row in the companion Lean 4 formalization; see Subsection 3.3 and Appendix A. The completeness of the geometric list is deduced later in Corollary 4.5 from the finite-field orbit-count identity. $\square$

In the quotient column, B , T , and N_T denote the full preimages in $\mathrm{GL}(L) \cong \mathrm{GL}_2(k)$ of a Borel subgroup, a maximal torus, and the normalizer of a maximal torus in $\mathrm{PGL}(L)$. The entry “ S_3 preimage” denotes the full preimage of $S_3 \leq \mathrm{PGL}(L)$.

3.1 Geometric orbit representatives

$n = 4$.

No.	Type	ω_1	ω_2
1	$J_{a,2}$	$e_1 \wedge e_3 + e_2 \wedge e_4$	$e_1 \wedge e_4$
2	$J_{a,1} + J_{b,1}$	$e_1 \wedge e_2 + e_3 \wedge e_4$	$e_3 \wedge e_4$
3	$S_1 + S_2$	$e_2 \wedge e_4$	$e_3 \wedge e_4$

$n = 5$.

No.	Type	ω_1	ω_2
1	S_3	$e_1 \wedge e_4 + e_2 \wedge e_5$	$e_2 \wedge e_4 + e_3 \wedge e_5$
2	$S_2 + S_1^2$	$e_1 \wedge e_3$	$e_2 \wedge e_3$
3	$S_2 + J_{a,1}$	$e_1 \wedge e_3 + e_4 \wedge e_5$	$e_2 \wedge e_3$
4	$S_1 + J_{a,2}$	$e_2 \wedge e_4 + e_3 \wedge e_5$	$e_2 \wedge e_5$
5	$S_1 + J_{a,1} + J_{b,1}$	$e_2 \wedge e_3 + e_4 \wedge e_5$	$e_4 \wedge e_5$

$n = 6$.

No.	Type	ω_1	ω_2
1	$J_{a,3}$	$e_1 \wedge e_4 + e_2 \wedge e_5 + e_3 \wedge e_6$	$e_1 \wedge e_5 + e_2 \wedge e_6$
2	$J_{a,2} + J_{a,1}$	$e_1 \wedge e_3 + e_2 \wedge e_4 + e_5 \wedge e_6$	$e_1 \wedge e_4$
3	$J_{a,2} + J_{b,1}$	$e_1 \wedge e_3 + e_2 \wedge e_4 + e_5 \wedge e_6$	$e_1 \wedge e_4 + e_5 \wedge e_6$
4	$2J_{a,1} + J_{b,1}$	$e_1 \wedge e_2 + e_3 \wedge e_4 + e_5 \wedge e_6$	$e_5 \wedge e_6$
5	$J_{a,1} + J_{b,1} + J_{c,1}$	$e_1 \wedge e_2 + e_3 \wedge e_4$	$e_3 \wedge e_4 + e_5 \wedge e_6$
6	$S_1^2 + J_{a,2}$	$e_3 \wedge e_5 + e_4 \wedge e_6$	$e_3 \wedge e_6$
7	$S_1^2 + J_{a,1} + J_{b,1}$	$e_3 \wedge e_4 + e_5 \wedge e_6$	$e_5 \wedge e_6$
8	$S_1 + S_2 + J_{a,1}$	$e_2 \wedge e_4 + e_5 \wedge e_6$	$e_3 \wedge e_4$
9	$S_1^3 + S_2$	$e_4 \wedge e_6$	$e_5 \wedge e_6$
10	$S_1 + S_3$	$e_2 \wedge e_5 + e_3 \wedge e_6$	$e_3 \wedge e_5 + e_4 \wedge e_6$
11	S_2^2	$e_1 \wedge e_3 + e_4 \wedge e_6$	$e_2 \wedge e_3 + e_5 \wedge e_6$

$n = 7$.

No.	Type	ω_1	ω_2
1	S_4	$e_1 \wedge e_5 + e_2 \wedge e_6 + e_3 \wedge e_7$	$e_2 \wedge e_5 + e_3 \wedge e_6 + e_4 \wedge e_7$
2	$S_3 + S_1^2$	$e_1 \wedge e_4 + e_2 \wedge e_5$	$e_2 \wedge e_4 + e_3 \wedge e_5$
3	$S_2^2 + S_1$	$e_1 \wedge e_3 + e_4 \wedge e_6$	$e_2 \wedge e_3 + e_5 \wedge e_6$
4	$S_2 + S_1^4$	$e_1 \wedge e_3$	$e_2 \wedge e_3$
5	$S_3 + J_{a,1}$	$e_1 \wedge e_4 + e_2 \wedge e_5 + e_6 \wedge e_7$	$e_2 \wedge e_4 + e_3 \wedge e_5$
6	$S_2 + S_1^2 + J_{a,1}$	$e_1 \wedge e_3 + e_6 \wedge e_7$	$e_2 \wedge e_3$
7	$S_2 + J_{a,2}$	$e_1 \wedge e_3 + e_4 \wedge e_6 + e_5 \wedge e_7$	$e_2 \wedge e_3 + e_4 \wedge e_7$
8	$S_2 + 2J_{a,1}$	$e_1 \wedge e_3 + e_4 \wedge e_5 + e_6 \wedge e_7$	$e_2 \wedge e_3$
9	$S_2 + J_{a,1} + J_{b,1}$	$e_1 \wedge e_3 + e_4 \wedge e_5 + e_6 \wedge e_7$	$e_2 \wedge e_3 + e_6 \wedge e_7$
10	$S_1^3 + J_{a,2}$	$e_4 \wedge e_6 + e_5 \wedge e_7$	$e_4 \wedge e_7$
11	$S_1^3 + J_{a,1} + J_{b,1}$	$e_4 \wedge e_5 + e_6 \wedge e_7$	$e_6 \wedge e_7$
12	$S_1 + J_{a,3}$	$e_2 \wedge e_5 + e_3 \wedge e_6 + e_4 \wedge e_7$	$e_2 \wedge e_6 + e_3 \wedge e_7$
13	$S_1 + J_{a,2} + J_{a,1}$	$e_2 \wedge e_4 + e_3 \wedge e_5 + e_6 \wedge e_7$	$e_2 \wedge e_5$
14	$S_1 + J_{a,2} + J_{b,1}$	$e_2 \wedge e_4 + e_3 \wedge e_5 + e_6 \wedge e_7$	$e_2 \wedge e_5 + e_6 \wedge e_7$
15	$S_1 + 2J_{a,1} + J_{b,1}$	$e_2 \wedge e_3 + e_4 \wedge e_5 + e_6 \wedge e_7$	$e_6 \wedge e_7$
16	$S_1 + J_{a,1} + J_{b,1} + J_{c,1}$	$e_2 \wedge e_3 + e_4 \wedge e_5$	$e_4 \wedge e_5 + e_6 \wedge e_7$

3.2 Geometric stabilizers

$n = 4$.

No.	Type	K_L in $\text{Stab}_{\text{GL}(V)}(L) = K_L \rtimes Q_L$	Q_L
1	$J_{a,2}$	$U_3 \rtimes \text{SL}_2(k)$	B
2	$J_{a,1} + J_{b,1}$	$\text{SL}_2(k) \times \text{SL}_2(k)$	N_T
3	$S_1 + S_2$	$U_5 \rtimes (\mathbf{G}_m \times \mathbf{G}_m)$	$\text{GL}_2(k)$

$n = 5$.

No.	Type	K_L in $\text{Stab}_{\text{GL}(V)}(L) = K_L \rtimes Q_L$	Q_L
1	S_3	$U_4 \rtimes \mathbf{G}_m$	$\text{GL}_2(k)$
2	$S_2 + S_1^2$	$U_8 \rtimes (\text{GL}_2(k) \times \mathbf{G}_m)$	$\text{GL}_2(k)$
3	$S_2 + J_{a,1}$	$U_4 \rtimes (\mathbf{G}_m \times \text{SL}_2(k))$	B
4	$S_1 + J_{a,2}$	$U_7 \rtimes (\mathbf{G}_m \times \text{SL}_2(k))$	B
5	$S_1 + J_{a,1} + J_{b,1}$	$U_4 \rtimes (\mathbf{G}_m \times \text{SL}_2(k) \times \text{SL}_2(k))$	N_T

$n = 6$.

No.	Type	K_L in $\text{Stab}_{\text{GL}(V)}(L) = K_L \rtimes Q_L$	Q_L
1	$J_{a,3}$	$U_6 \rtimes \text{SL}_2(k)$	B
2	$J_{a,2} + J_{a,1}$	$U_7 \rtimes (\text{SL}_2(k) \times \text{SL}_2(k))$	B
3	$J_{a,2} + J_{b,1}$	$U_3 \rtimes (\text{SL}_2(k) \times \text{SL}_2(k))$	T
4	$2J_{a,1} + J_{b,1}$	$\text{Sp}_4(k) \times \text{SL}_2(k)$	T
5	$J_{a,1} + J_{b,1} + J_{c,1}$	$\text{SL}_2(k) \times \text{SL}_2(k) \times \text{SL}_2(k)$	S_3 preimage
6	$S_1^2 + J_{a,2}$	$U_{11} \rtimes (\text{GL}_2(k) \times \text{SL}_2(k))$	B
7	$S_1^2 + J_{a,1} + J_{b,1}$	$U_8 \rtimes (\text{GL}_2(k) \times \text{SL}_2(k) \times \text{SL}_2(k))$	N_T
8	$S_1 + S_2 + J_{a,1}$	$U_9 \rtimes (\mathbf{G}_m \times \mathbf{G}_m \times \text{SL}_2(k))$	B

Continued on next page

No.	Type	K_L in $\text{Stab}_{\text{GL}(V)}(L) = K_L \rtimes Q_L$	Q_L
9	$S_1^3 + S_2$	$U_{11} \rtimes (\text{GL}_3(k) \times \mathbf{G}_m)$	$\text{GL}_2(k)$
10	$S_1 + S_3$	$U_8 \rtimes (\mathbf{G}_m \times \mathbf{G}_m)$	$\text{GL}_2(k)$
11	S_2^2	$U_6 \rtimes \text{GL}_2(k)$	$\text{GL}_2(k)$

$n = 7$.

No.	Type	K_L in $\text{Stab}_{\text{GL}(V)}(L) = K_L \rtimes Q_L$	Q_L
1	S_4	$\mathbf{G}_a^{14} \rtimes (\mathbf{G}_m \times \mathbf{G}_m)$	$\text{GL}_2(k)$
2	$S_3 + S_1^2$	$\mathbf{G}_a^{10} \rtimes (\mathbf{G}_m \times (U_4 \rtimes \mathbf{G}_m))$	$\text{GL}_2(k)$
3	$S_2^2 + S_1$	$\mathbf{G}_a^6 \rtimes (\mathbf{G}_m \times (U_6 \rtimes \text{GL}_2(k)))$	$\text{GL}_2(k)$
4	$S_2 + S_1^4$	$\mathbf{G}_a^{12} \rtimes (\text{GL}_4(k) \times (\mathbf{G}_a^2 \rtimes \mathbf{G}_m))$	$\text{GL}_2(k)$
5	$S_3 + J_{a,1}$	$\mathbf{G}_a^{10} \rtimes (\text{SL}_2(k) \times (U_4 \rtimes \mathbf{G}_m))$	B
6	$S_2 + S_1^2 + J_{a,1}$	$\mathbf{G}_a^{10} \rtimes (\mathbf{G}_m \times (U_4 \rtimes (\mathbf{G}_m \times \text{SL}_2(k))))$	B
7	$S_2 + J_{a,2}$	$\mathbf{G}_a^6 \rtimes (\mathbf{G}_m \times (U_3 \rtimes \text{SL}_2(k)))$	B
8	$S_2 + 2J_{a,1}$	$\mathbf{G}_a^6 \rtimes (\mathbf{G}_m \times \text{Sp}_4(k))$	B
9	$S_2 + J_{a,1} + J_{b,1}$	$\mathbf{G}_a^6 \rtimes (\mathbf{G}_m \times (\text{SL}_2(k) \times \text{SL}_2(k)))$	N_T
10	$S_1^3 + J_{a,2}$	$\mathbf{G}_a^{12} \rtimes (\text{GL}_3(k) \times \text{SL}_2(k[t]/(t^2)))$	B
11	$S_1^3 + J_{a,1} + J_{b,1}$	$\mathbf{G}_a^{12} \rtimes (\text{GL}_3(k) \times (\text{SL}_2(k) \times \text{SL}_2(k)))$	N_T
12	$S_1 + J_{a,3}$	$\mathbf{G}_a^6 \rtimes (\mathbf{G}_m \times \text{SL}_2(k[t]/(t^3)))$	B
13	$S_1 + J_{a,2} + J_{a,1}$	$\mathbf{G}_a^6 \rtimes (\mathbf{G}_m \times (U_{2,1}(k) \rtimes (\text{SL}_2(k) \times \text{SL}_2(k))))$	B
14	$S_1 + J_{a,2} + J_{b,1}$	$\mathbf{G}_a^6 \rtimes (\mathbf{G}_m \times (\text{SL}_2(k) \times \text{SL}_2(k)))$	T
15	$S_1 + 2J_{a,1} + J_{b,1}$	$\mathbf{G}_a^6 \rtimes (\mathbf{G}_m \times (\text{Sp}_4(k) \times \text{SL}_2(k)))$	T
16	$S_1 + J_{a,1} + J_{b,1} + J_{c,1}$	$\mathbf{G}_a^6 \rtimes (\mathbf{G}_m \times (\text{SL}_2(k) \times \text{SL}_2(k) \times \text{SL}_2(k)))$	S_3 preimage

3.3 How the formalization is used

For each table row there is a public Lean namespace `Wedge2Formalization.Paper.Nn.Rowm` and a matching source file `Wedge2Formalization/Paper/Nn/Rowm.lean` in the companion repository. This is the referee-facing layer of the formalization; the lower-level coordinate files are not needed for a row-by-row check.

Inside that namespace, the declarations `paperRep1` and `paperRep2` are the literal basis vectors printed in the geometric orbit table. When the proof uses a more convenient internal working model, the lemmas `paperRep1_transport` and `paperRep2_transport` identify the displayed representative with that internal model.

The theorem `mem_K_table_iff` is the public kernel statement for the row, and the theorem `quotient_image` identifies the image Q_L of the coefficient-side action. Together with the representative transport lemmas, these declarations recover the stabilizer line printed in the tables. Appendix A records the row-to-namespace correspondence.

4 Finite-Field Consequences

Now let $V = \mathbf{F}_q^n$. The weak-congruence row labels from Section 3 remain in force, but over $\mathbf{F}_q$ some geometric rows split according to the Frobenius action on the underlying simple support factors. The resulting finite-field stabilizers are recorded in the tables below.

Proposition 4.1. *Let L be one of the representatives defined over $\mathbf{F}_q$, and let*

$$H_L = \text{Stab}_{\text{GL}_n(\overline{\mathbf{F}}_q)}(L).$$

Then

$$\text{Stab}_{\text{GL}_n(q)}(L) = H_L^F,$$

where F is the standard Frobenius endomorphism.

Proof. Since L is F -stable, the stabilizer over $\mathbf{F}_q$ is the fixed-point group of the algebraic stabilizer. Indeed,

$$g \in \text{GL}_n(q) \iff F(g) = g.$$

If such a g stabilizes L , then it lies in H_L^F . Conversely every element of H_L^F is an F -fixed element of $\text{GL}_n(\overline{\mathbf{F}}_q)$ that stabilizes L , hence belongs to $\text{GL}_n(q)$ and stabilizes L over $\mathbf{F}_q$. Therefore

$$\text{Stab}_{\text{GL}_n(q)}(L) = H_L^F.$$

□

Proposition 4.2. *Every finite-field stabilizer through $n \leq 7$ is obtained from the corresponding algebraic stabilizer by Frobenius fixed points. Distinct support points in one Frobenius orbit give the field-extension groups $\text{SL}_2(q^2)$ and $\text{SL}_2(q^3)$. Repeated support at one rational point gives the groups $U_{3d-3}(q) \rtimes \text{SL}_2(q)$.*

The orbit-by-orbit formulas are listed in the finite-field tables below.

Proof. Let L be an $\mathbf{F}_q$ -rational representative. Proposition 4.1 identifies its finite-field stabilizer with the fixed-point group of the algebraic stabilizer. We therefore only need to understand how Frobenius acts on the quotient Q_L and on the kernel K_L .

On the quotient side, rational support gives the split families already visible over k . If two or three simple-support factors form one Frobenius orbit, then the induced permutation on the coefficient line produces the non-split torus normalizer or the cyclic quadratic/cubic quotient recorded in the finite-field tables.

On the kernel side, there are two cases. If the support points are distinct and lie in one Frobenius orbit, the corresponding simple $\text{SL}_2(k)$ -factors are identified by Frobenius, and the fixed points are the field-extension groups $\text{SL}_2(q^2)$ or $\text{SL}_2(q^3)$. If one support point is repeated, then the intersection calculation already lives over the rational point, so Frobenius acts coefficientwise on the parameters appearing in the unipotent radical and on the Levi factor $\text{SL}_2(k)$. The fixed-point group therefore has unipotent radical of dimension $3d - 3$ and Levi quotient $\text{SL}_2(q)$. Therefore it is written in the tables as $U_{3d-3}(q) \rtimes \text{SL}_2(q)$. Radical extensions descend termwise because their defining formulas are polynomial in the coefficients of the representative. The same is true for the one-flat extensions from the geometric table. □

Here $\lambda, \mu, \nu \in \mathbf{F}_q$ denote distinct rational parameters. The symbol Q denotes the quadratic simple-support row, and C denotes the cubic simple-support row. In the quotient column, $B(q)$, $T_s(q)$, $N_s(q)$, and $N_{ns}(q)$ denote the split Borel subgroup of $\text{GL}_2(q)$, the split maximal torus, the normalizer of the split torus, and the normalizer of a non-split torus. The groups $\tilde{S}_3(q)$, $\tilde{C}_2(q)$, and $\tilde{C}_3(q)$ are the corresponding full preimages in $\text{GL}_2(q)$ of S_3 , C_2 , and C_3 in $\text{PGL}_2(q)$.

$n = 4.$

No.	Type	K_L in $\text{Stab}_{\text{GL}_n(q)}(L) = K_L \rtimes Q_L$	Q_L
1	$J_{\lambda,2}$	$U_3(q) \rtimes \text{SL}_2(q)$	$B(q)$
2	$J_{\lambda,1} + J_{\mu,1} \ (\lambda \neq \mu)$	$\text{SL}_2(q) \times \text{SL}_2(q)$	$N_s(q)$
3	Q	$\text{SL}_2(q^2)$	$N_{ns}(q)$
4	$S_1 + S_2$	$U_5(q) \rtimes (\text{GL}_1(q) \times \text{GL}_1(q))$	$\text{GL}_2(q)$

$n = 5.$

No.	Type	K_L in $\text{Stab}_{\text{GL}_n(q)}(L) = K_L \rtimes Q_L$	Q_L
1	S_3	$U_4(q) \rtimes \text{GL}_1(q)$	$\text{GL}_2(q)$
2	$S_2 + S_1^2$	$U_8(q) \rtimes (\text{GL}_2(q) \times \text{GL}_1(q))$	$\text{GL}_2(q)$
3	$S_2 + J_{\lambda,1}$	$U_4(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q))$	$B(q)$
4	$S_1 + J_{\lambda,2}$	$U_7(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q))$	$B(q)$
5	$S_1 + J_{\lambda,1} + J_{\mu,1} \ (\lambda \neq \mu)$	$U_4(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q) \times \text{SL}_2(q))$	$N_s(q)$
6	$S_1 + Q$	$U_4(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q^2))$	$N_{ns}(q)$

$n = 6.$

No.	Type	K_L in $\text{Stab}_{\text{GL}_n(q)}(L) = K_L \rtimes Q_L$	Q_L
1	$J_{\lambda,3}$	$U_6(q) \rtimes \text{SL}_2(q)$	$B(q)$
2	$J_{\lambda,2} + J_{\lambda,1}$	$U_7(q) \rtimes (\text{SL}_2(q) \times \text{SL}_2(q))$	$B(q)$
3	$J_{\lambda,2} + J_{\mu,1} \ (\lambda \neq \mu)$	$U_3(q) \rtimes (\text{SL}_2(q) \times \text{SL}_2(q))$	$T_s(q)$
4	$2J_{\lambda,1} + J_{\mu,1} \ (\lambda \neq \mu)$	$\text{Sp}_4(q) \times \text{SL}_2(q)$	$T_s(q)$
5	$J_{\lambda,1} + J_{\mu,1} + J_{\nu,1} \ (\text{distinct})$	$\text{SL}_2(q) \times \text{SL}_2(q) \times \text{SL}_2(q)$	$\tilde{S}_3(q)$
6	$J_{\lambda,1} + Q$	$\text{SL}_2(q) \times \text{SL}_2(q^2)$	$\tilde{C}_2(q)$
7	C	$\text{SL}_2(q^3)$	$\tilde{C}_3(q)$
8	$S_1^2 + J_{\lambda,2}$	$U_{11}(q) \rtimes (\text{GL}_2(q) \times \text{SL}_2(q))$	$B(q)$
9	$S_1^2 + J_{\lambda,1} + J_{\mu,1} \ (\lambda \neq \mu)$	$U_8(q) \rtimes (\text{GL}_2(q) \times \text{SL}_2(q) \times \text{SL}_2(q))$	$N_s(q)$
10	$S_1^2 + Q$	$U_8(q) \rtimes (\text{GL}_2(q) \times \text{SL}_2(q^2))$	$N_{ns}(q)$
11	$S_1 + S_2 + J_{\lambda,1}$	$U_9(q) \rtimes (\text{GL}_1(q) \times \text{GL}_1(q) \times \text{SL}_2(q))$	$B(q)$
12	$S_1^3 + S_2$	$U_{11}(q) \rtimes (\text{GL}_3(q) \times \text{GL}_1(q))$	$\text{GL}_2(q)$
13	$S_1 + S_3$	$U_8(q) \rtimes (\text{GL}_1(q) \times \text{GL}_1(q))$	$\text{GL}_2(q)$
14	S_2^2	$U_6(q) \rtimes \text{GL}_2(q)$	$\text{GL}_2(q)$

$n = 7.$

No.	Type	K_L in $\text{Stab}_{\text{GL}_n(q)}(L) = K_L \rtimes Q_L$	Q_L
1	S_4	$U_6(q) \rtimes \text{GL}_1(q)$	$\text{GL}_2(q)$
2	$S_3 + S_1^2$	$U_{14}(q) \rtimes (\text{GL}_2(q) \times \text{GL}_1(q))$	$\text{GL}_2(q)$
3	$S_2^2 + S_1$	$U_{12}(q) \rtimes (\text{GL}_1(q) \times \text{GL}_2(q))$	$\text{GL}_2(q)$
4	$S_2 + S_1^4$	$U_{14}(q) \rtimes (\text{GL}_4(q) \times \text{GL}_1(q))$	$\text{GL}_2(q)$
5	$S_3 + J_{\lambda,1}$	$U_6(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q))$	$B(q)$
6	$S_2 + S_1^2 + J_{\lambda,1}$	$U_{14}(q) \rtimes (\text{GL}_2(q) \times \text{GL}_1(q) \times \text{SL}_2(q))$	$B(q)$
7	$S_2 + J_{\lambda,2}$	$U_9(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q))$	$B(q)$
8	$S_2 + 2J_{\lambda,1}$	$U_6(q) \rtimes (\text{GL}_1(q) \times \text{Sp}_4(q))$	$B(q)$
9	$S_2 + J_{\lambda,1} + J_{\mu,1} \ (\lambda \neq \mu)$	$U_6(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q) \times \text{SL}_2(q))$	$N_s(q)$
10	$S_2 + Q$	$U_6(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q^2))$	$N_{ns}(q)$
11	$S_1^3 + J_{\lambda,2}$	$U_{15}(q) \rtimes (\text{GL}_3(q) \times \text{SL}_2(q))$	$B(q)$
12	$S_1^3 + J_{\lambda,1} + J_{\mu,1} \ (\lambda \neq \mu)$	$U_{12}(q) \rtimes (\text{GL}_3(q) \times \text{SL}_2(q) \times \text{SL}_2(q))$	$N_s(q)$
13	$S_1^3 + Q$	$U_{12}(q) \rtimes (\text{GL}_3(q) \times \text{SL}_2(q^2))$	$N_{ns}(q)$

Continued on next page

No.	Type	K_L in $\text{Stab}_{\text{GL}_n(q)}(L) = K_L \rtimes Q_L$	Q_L
14	$S_1 + J_{\lambda,3}$	$U_{12}(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q))$	$B(q)$
15	$S_1 + J_{\lambda,2} + J_{\lambda,1}$	$U_{13}(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q) \times \text{SL}_2(q))$	$B(q)$
16	$S_1 + J_{\lambda,2} + J_{\mu,1} \ (\lambda \neq \mu)$	$U_9(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q) \times \text{SL}_2(q))$	$T_s(q)$
17	$S_1 + 2J_{\lambda,1} + J_{\mu,1} \ (\lambda \neq \mu)$	$U_6(q) \rtimes (\text{GL}_1(q) \times \text{Sp}_4(q) \times \text{SL}_2(q))$	$T_s(q)$
18	$S_1 + J_{\lambda,1} + J_{\mu,1} + J_{\nu,1} \text{ (distinct)}$	$U_6(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q) \times \text{SL}_2(q) \times \text{SL}_2(q))$	$\tilde{S}_3(q)$
19	$S_1 + J_{\lambda,1} + Q$	$U_6(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q) \times \text{SL}_2(q^2))$	$\tilde{C}_2(q)$
20	$S_1 + C$	$U_6(q) \rtimes (\text{GL}_1(q) \times \text{SL}_2(q^3))$	$\tilde{C}_3(q)$

Theorem 4.3. *For each $n = 4, 5, 6, 7$, let $\mathcal{T}_n(q)$ be the finite list of orbit representatives in the finite-field tables above. Then*

$$\sum_{L \in \mathcal{T}_n(q)} \frac{|\text{GL}_n(q)|}{|\text{Stab}_{\text{GL}_n(q)}(L)|} = \left[\begin{matrix} n \\ 2 \end{matrix} \right]_q.$$

Therefore the finite-field tables above are the complete orbit table for $\text{GL}_n(q)$ on $\text{Gr}(2, \wedge^2 \mathbf{F}_q^n)$.

Proof. The right-hand side is the number of 2-spaces in $\wedge^2 \mathbf{F}_q^n$. The left-hand side is the total number of elements covered by the listed orbits. For each row of the finite-field tables, the orbit-stabilizer theorem gives an orbit of size

$$\frac{|\text{GL}_n(q)|}{|\text{Stab}_{\text{GL}_n(q)}(L)|}.$$

Summing over all listed representatives therefore counts the total number of points lying in the listed orbits.

It remains to compare this sum with the number of all 2-spaces in $\wedge^2 \mathbf{F}_q^n$, which is the Gaussian polynomial on the right. Substituting the explicit stabilizer formulas from the finite-field tables gives a rational function in q . After clearing denominators, one obtains a polynomial identity. The symbolic simplification is routine but lengthy, and we verify it with the companion Magma scripts archived with the paper. Therefore the listed orbit sizes add up to the whole Grassmannian.

Since distinct orbits are disjoint, no further orbit can exist. Therefore the finite-field tables are complete. $\square$

Corollary 4.4. *The numbers of $\text{GL}_n(q)$ -orbits on $\text{Gr}(2, \wedge^2 \mathbf{F}_q^n)$ are*

$$4, 6, 14, 20$$

for $n = 4, 5, 6, 7$, respectively.

Proof. The finite-field tables above have exactly 4, 6, 14, and 20 rows for $n = 4, 5, 6$, and 7, respectively. $\square$

Corollary 4.5. *The geometric orbit representatives in Section 3 form a complete set of $\text{GL}(V)$ -orbit representatives on $\text{Gr}(2, \wedge^2 V)$ for $\dim V = 4, 5, 6, 7$. The numbers of geometric orbits are*

$$3, 5, 11, 16$$

for $\dim V = 4, 5, 6, 7$, respectively.

Proof. Let $k = \overline{\mathbf{F}_q}$, let $V = k^n$ with $n \leq 7$, and let $L \leq \wedge^2 V$ be any geometric 2-space. Choose a basis ω_1, ω_2 of L . Since k is the algebraic closure of $\mathbf{F}_q$, the finitely many coefficients of ω_1 and ω_2 lie in some finite extension $\mathbf{F}_{q^m} \subset k$. Therefore L is defined over $\mathbf{F}_{q^m}$.

Apply Theorem 4.3 with q replaced by q^m . The resulting finite-field orbit table is complete for $\mathrm{GL}_n(q^m)$ on $\mathrm{Gr}(2, \wedge^2 \mathbf{F}_{q^m}^n)$. Hence L is $\mathrm{GL}_n(q^m)$ -equivalent to one of the finite-field representatives listed in this section. Each of those finite-field representatives is obtained by Frobenius descent from one of the geometric representatives in Section 3. After extending scalars back to k , the same equivalence holds in $\mathrm{GL}_n(k)$. Therefore every geometric orbit is represented by one of the listed geometric rows.

Counting the rows in Section 3 gives 3, 5, 11, and 16 representatives in dimensions 4, 5, 6, and 7, respectively. $\square$

A Weak-congruence labels and Lean crosswalk

The public Lean entry point is `Wedge2Formalization.Paper`. For each row of the geometric tables there is a corresponding namespace `Wedge2Formalization.Paper.Nn.Rowm` and a dedicated file `Wedge2Formalization/Paper/Nn/Rowm.lean`. In every public row namespace, the representative transport is given by `paperRep1_transport` and `paperRep2_transport`, the table-facing kernel theorem is `mem_K_table_iff`, and the quotient theorem is `quotient_image`.

The following tables record the row numbering used in the paper, the weak-congruence label used to name the representative, and the corresponding public Lean namespace.

$n = 4$.

Row	Type	Lean namespace
1	$J_{a,2}$	<code>Wedge2Formalization.Paper.N4.Row1</code>
2	$J_{a,1} + J_{b,1}$	<code>Wedge2Formalization.Paper.N4.Row2</code>
3	$S_1 + S_2$	<code>Wedge2Formalization.Paper.N4.Row3</code>

$n = 5$.

Row	Type	Lean namespace
1	S_3	<code>Wedge2Formalization.Paper.N5.Row1</code>
2	$S_2 + S_1^2$	<code>Wedge2Formalization.Paper.N5.Row2</code>
3	$S_2 + J_{a,1}$	<code>Wedge2Formalization.Paper.N5.Row3</code>
4	$S_1 + J_{a,2}$	<code>Wedge2Formalization.Paper.N5.Row4</code>
5	$S_1 + J_{a,1} + J_{b,1}$	<code>Wedge2Formalization.Paper.N5.Row5</code>

$n = 6$.

Row	Type	Lean namespace
1	$J_{a,3}$	<code>Wedge2Formalization.Paper.N6.Row1</code>
2	$J_{a,2} + J_{a,1}$	<code>Wedge2Formalization.Paper.N6.Row2</code>
3	$J_{a,2} + J_{b,1}$	<code>Wedge2Formalization.Paper.N6.Row3</code>
4	$2J_{a,1} + J_{b,1}$	<code>Wedge2Formalization.Paper.N6.Row4</code>
5	$J_{a,1} + J_{b,1} + J_{c,1}$	<code>Wedge2Formalization.Paper.N6.Row5</code>
6	$S_1^2 + J_{a,2}$	<code>Wedge2Formalization.Paper.N6.Row6</code>
7	$S_1^2 + J_{a,1} + J_{b,1}$	<code>Wedge2Formalization.Paper.N6.Row7</code>
8	$S_1 + S_2 + J_{a,1}$	<code>Wedge2Formalization.Paper.N6.Row8</code>
9	$S_1^3 + S_2$	<code>Wedge2Formalization.Paper.N6.Row9</code>
10	$S_1 + S_3$	<code>Wedge2Formalization.Paper.N6.Row10</code>
11	S_2^2	<code>Wedge2Formalization.Paper.N6.Row11</code>

$n = 7$.

Row	Type	Lean namespace
1	S_4	<code>Wedge2Formalization.Paper.N7.Row1</code>
2	$S_3 + S_1^2$	<code>Wedge2Formalization.Paper.N7.Row2</code>
3	$S_2^2 + S_1$	<code>Wedge2Formalization.Paper.N7.Row3</code>
4	$S_2 + S_1^4$	<code>Wedge2Formalization.Paper.N7.Row4</code>
5	$S_3 + J_{a,1}$	<code>Wedge2Formalization.Paper.N7.Row5</code>
6	$S_2 + S_1^2 + J_{a,1}$	<code>Wedge2Formalization.Paper.N7.Row6</code>
7	$S_2 + J_{a,2}$	<code>Wedge2Formalization.Paper.N7.Row7</code>
8	$S_2 + 2J_{a,1}$	<code>Wedge2Formalization.Paper.N7.Row8</code>
9	$S_2 + J_{a,1} + J_{b,1}$	<code>Wedge2Formalization.Paper.N7.Row9</code>
10	$S_1^3 + J_{a,2}$	<code>Wedge2Formalization.Paper.N7.Row10</code>
11	$S_1^3 + J_{a,1} + J_{b,1}$	<code>Wedge2Formalization.Paper.N7.Row11</code>
12	$S_1 + J_{a,3}$	<code>Wedge2Formalization.Paper.N7.Row12</code>
13	$S_1 + J_{a,2} + J_{a,1}$	<code>Wedge2Formalization.Paper.N7.Row13</code>
14	$S_1 + J_{a,2} + J_{b,1}$	<code>Wedge2Formalization.Paper.N7.Row14</code>
15	$S_1 + 2J_{a,1} + J_{b,1}$	<code>Wedge2Formalization.Paper.N7.Row15</code>
16	$S_1 + J_{a,1} + J_{b,1} + J_{c,1}$	<code>Wedge2Formalization.Paper.N7.Row16</code>

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